In [103…
```
'''Name of Student: Mansi Mukund Kondhare
PRN No : 72332141J
Class & Div:B E Div-B
Roll No: 26'''
```

Out[103…
```
'Name of Student: Mansi Mukund Kondhare\nPRN No : 72332141J \nClass & Di
v:B E Div-B \nRoll No: 26'
```

In [104…
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [105…
```
# 1. LOAD CSV FILE
df=pd.read_csv('sales.csv', encoding='latin-1')
```

In [106…
```
df
```

Out[106…

	ORDER_ID	ORDER_DATE	CUSTOMER	CITY	COUNTRY	PRODUCT	CATEG
0	1001	2026-01-05	Amit	Pune	India	Laptop	Electro
1	1002	2026-01-08	Priya	Mumbai	India	Mouse	Electro
2	1003	2026-01-12	Rahul	Delhi	India	Office Chair	Furni
3	1004	2026-01-15	Sneha	Pune	India	Desk	Furni
4	1005	2026-01-20	Amit	Bengaluru	India	Keyboard	Electro
5	1006	2026-01-25	Priya	Mumbai	India	Monitor	Electro
6	1007	2026-02-02	Rahul	Delhi	India	Bookshelf	Furni
7	1008	2026-02-06	Neha	Pune	India	Tablet	Electro
8	1009	2026-02-10	Raj	Bengaluru	India	Sofa	Furni
9	1010	2026-02-14	Pooja	Mumbai	India	Headphones	Electro
10	1011	2026-02-18	Karan	Delhi	India	Printer	Electro
11	1012	2026-02-22	Riya	Pune	India	Cabinet	Furni
12	1013	2026-03-01	Arjun	Bengaluru	India	Laptop	Electro
13	1014	2026-03-05	Meera	Mumbai	India	Desk	Furni
14	1015	2026-03-10	Vikas	Delhi	India	Mouse	Electro
15	1016	2026-03-15	Anjali	Pune	India	Office Chair	Furni
16	1017	2026-03-20	Rohit	Bengaluru	India	Monitor	Electro
17	1018	2026-03-24	Kavya	Mumbai	India	Table	Furni
18	1019	2026-03-27	Sahil	Delhi	India	Keyboard	Electro
19	1020	2026-03-30	Nisha	Pune	India	Tablet	Electro
20	1021	2026-04-03	Varun	Mumbai	India	Printer	Electro
21	1022	2026-04-07	Isha	Delhi	India	Chair	Furni
22	1023	2026-04-10	Aarav	Pune	India	Laptop	Electro
23	1023	2026-04-10	Aarav	Pune	India	Laptop	Electro

In [107…

```
# 2. DISPLAY FIRST 5 RECORDS
df.head()
```

Out[107…

	ORDER_ID	ORDER_DATE	CUSTOMER	CITY	COUNTRY	PRODUCT	CATEGORY
0	1001	2026-01-05	Amit	Pune	India	Laptop	Electronics
1	1002	2026-01-08	Priya	Mumbai	India	Mouse	Electronics
2	1003	2026-01-12	Rahul	Delhi	India	Office Chair	Furniture
3	1004	2026-01-15	Sneha	Pune	India	Desk	Furniture
4	1005	2026-01-20	Amit	Bengaluru	India	Keyboard	Electronics

In [108…
```
# 3. DISPLAY LAST 5 RECORDS
df.tail()
```

Out[108…

	ORDER_ID	ORDER_DATE	CUSTOMER	CITY	COUNTRY	PRODUCT	CATEGORY
19	1020	2026-03-30	Nisha	Pune	India	Tablet	Electronics
20	1021	2026-04-03	Varun	Mumbai	India	Printer	Electronics
21	1022	2026-04-07	Isha	Delhi	India	Chair	Furniture
22	1023	2026-04-10	Aarav	Pune	India	Laptop	Electronics
23	1023	2026-04-10	Aarav	Pune	India	Laptop	Electronics

In [109…
```
# 4. CHECK SHAPE OF DATA
df.shape
```

Out[109… (24, 9)

In [110…
```
# 5. CHECK INFORMATION
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24 entries, 0 to 23
Data columns (total 9 columns):
 #   Column      Non-Null Count  Dtype  
---  ------      --------------  -----  
 0   ORDER_ID    24 non-null     int64  
 1   ORDER_DATE  24 non-null     object 
 2   CUSTOMER    24 non-null     object 
 3   CITY        24 non-null     object 
 4   COUNTRY     24 non-null     object 
 5   PRODUCT     24 non-null     object 
 6   CATEGORY    24 non-null     object 
 7   QUANTITY    23 non-null     float64
 8   PRICE       23 non-null     float64
dtypes: float64(2), int64(1), object(6)
memory usage: 1.8+ KB
```

In [111…
```
# 6. CHECK MISSING VALUES
df.isnull().sum()
```

Out[111…

	0
ORDER_ID	0
ORDER_DATE	0
CUSTOMER	0
CITY	0
COUNTRY	0
PRODUCT	0
CATEGORY	0
QUANTITY	1
PRICE	1

dtype: int64

In [112…

```
# 7. DESCRIPTIVE STATISTICS
df.describe()
```

Out[112…

	ORDER_ID	QUANTITY	PRICE
count	24.000000	23.000000	23.000000
mean	1012.458333	2.043478	17984.782609
std	7.002976	1.364427	20064.053309
min	1001.000000	1.000000	650.000000
25%	1006.750000	1.000000	4350.000000
50%	1012.500000	2.000000	9500.000000
75%	1018.250000	2.000000	23000.000000
max	1023.000000	6.000000	60000.000000

In [113…

```
# 8. HANDLE MISSING VALUES
newdf = df.fillna(0)
df.isnull().sum()
```

Out[113…

	0
ORDER_ID	0
ORDER_DATE	0
CUSTOMER	0
CITY	0
COUNTRY	0
PRODUCT	0
CATEGORY	0
QUANTITY	1
PRICE	1

dtype: int64

In [114…

```
# 9. DISPLAY CLEANED DATA
newdf
```

Out[114…

	ORDER_ID	ORDER_DATE	CUSTOMER	CITY	COUNTRY	PRODUCT	CATEG
0	1001	2026-01-05	Amit	Pune	India	Laptop	Electro
1	1002	2026-01-08	Priya	Mumbai	India	Mouse	Electro
2	1003	2026-01-12	Rahul	Delhi	India	Office Chair	Furni
3	1004	2026-01-15	Sneha	Pune	India	Desk	Furni
4	1005	2026-01-20	Amit	Bengaluru	India	Keyboard	Electro
5	1006	2026-01-25	Priya	Mumbai	India	Monitor	Electro
6	1007	2026-02-02	Rahul	Delhi	India	Bookshelf	Furni
7	1008	2026-02-06	Neha	Pune	India	Tablet	Electro
8	1009	2026-02-10	Raj	Bengaluru	India	Sofa	Furni
9	1010	2026-02-14	Pooja	Mumbai	India	Headphones	Electro
10	1011	2026-02-18	Karan	Delhi	India	Printer	Electro
11	1012	2026-02-22	Riya	Pune	India	Cabinet	Furni
12	1013	2026-03-01	Arjun	Bengaluru	India	Laptop	Electro
13	1014	2026-03-05	Meera	Mumbai	India	Desk	Furni
14	1015	2026-03-10	Vikas	Delhi	India	Mouse	Electro
15	1016	2026-03-15	Anjali	Pune	India	Office Chair	Furni
16	1017	2026-03-20	Rohit	Bengaluru	India	Monitor	Electro
17	1018	2026-03-24	Kavya	Mumbai	India	Table	Furni
18	1019	2026-03-27	Sahil	Delhi	India	Keyboard	Electro
19	1020	2026-03-30	Nisha	Pune	India	Tablet	Electro
20	1021	2026-04-03	Varun	Mumbai	India	Printer	Electro
21	1022	2026-04-07	Isha	Delhi	India	Chair	Furni
22	1023	2026-04-10	Aarav	Pune	India	Laptop	Electro
23	1023	2026-04-10	Aarav	Pune	India	Laptop	Electro

In [115…

```
# 10. CREATE NEW LOCATION COLUMN
newdf['LOCATION'] = newdf['CITY'] + ',' + newdf['COUNTRY']

newdf.head()
```

Out[115…

	ORDER_ID	ORDER_DATE	CUSTOMER	CITY	COUNTRY	PRODUCT	CATEGORY
0	1001	2026-01-05	Amit	Pune	India	Laptop	Electronics
1	1002	2026-01-08	Priya	Mumbai	India	Mouse	Electronics
2	1003	2026-01-12	Rahul	Delhi	India	Office Chair	Furniture
3	1004	2026-01-15	Sneha	Pune	India	Desk	Furniture
4	1005	2026-01-20	Amit	Bengaluru	India	Keyboard	Electronics

In [116…
```
## 11. CHECK COLUMN NAMES
newdf.columns
```

Out[116…
```
Index(['ORDER_ID', 'ORDER_DATE', 'CUSTOMER', 'CITY', 'COUNTRY', 'PRODUCT',
       'CATEGORY', 'QUANTITY', 'PRICE', 'LOCATION'],
      dtype='object')
```

In [117…
```
# 12. CHECK DUPLICATE VALUES
newdf.duplicated().sum()
```

Out[117…
```
np.int64(1)
```

In [118…
```
# 13. REMOVE DUPLICATE ROWS
newdf = newdf.drop_duplicates()

newdf.shape
```

Out[118…
```
(23, 10)
```

In [119…
```
## 14. LOAD EXCEL FILE
excel_df = pd.read_excel('sales.xlsx')

excel_df
```

Out[119…

	ORDER_ID	ORDER_DATE	CUSTOMER	CITY	COUNTRY	PRODUCT	CATEG
0	1001	2026-01-05	Amit	Pune	India	Laptop	Electro
1	1002	2026-01-08	Priya	Mumbai	India	Mouse	Electro
2	1003	2026-01-12	Rahul	Delhi	India	Office Chair	Furni
3	1004	2026-01-15	Sneha	Pune	India	Desk	Furni
4	1005	2026-01-20	Amit	Bengaluru	India	Keyboard	Electro
5	1006	2026-01-25	Priya	Mumbai	India	Monitor	Electro
6	1007	2026-02-02	Rahul	Delhi	India	Bookshelf	Furni
7	1008	2026-02-06	Neha	Pune	India	Tablet	Electro
8	1009	2026-02-10	Raj	Bengaluru	India	Sofa	Furni
9	1010	2026-02-14	Pooja	Mumbai	India	Headphones	Electro
10	1011	2026-02-18	Karan	Delhi	India	Printer	Electro
11	1012	2026-02-22	Riya	Pune	India	Cabinet	Furni
12	1013	2026-03-01	Arjun	Bengaluru	India	Laptop	Electro
13	1014	2026-03-05	Meera	Mumbai	India	Desk	Furni
14	1015	2026-03-10	Vikas	Delhi	India	Mouse	Electro
15	1016	2026-03-15	Anjali	Pune	India	Office Chair	Furni
16	1017	2026-03-20	Rohit	Bengaluru	India	Monitor	Electro
17	1018	2026-03-24	Kavya	Mumbai	India	Table	Furni
18	1019	2026-03-27	Sahil	Delhi	India	Keyboard	Electro
19	1020	2026-03-30	Nisha	Pune	India	Tablet	Electro
20	1021	2026-04-03	Varun	Mumbai	India	Printer	Electro
21	1022	2026-04-07	Isha	Delhi	India	Chair	Furni
22	1023	2026-04-10	Aarav	Pune	India	Laptop	Electro
23	1023	2026-04-10	Aarav	Pune	India	Laptop	Electro

In [120…

```
#15. DISPLAY FIRST 5 EXCEL RECORDS
excel_df.head()
```

Out[120…

	ORDER_ID	ORDER_DATE	CUSTOMER	CITY	COUNTRY	PRODUCT	CATEGORY
0	1001	2026-01-05	Amit	Pune	India	Laptop	Electronics
1	1002	2026-01-08	Priya	Mumbai	India	Mouse	Electronics
2	1003	2026-01-12	Rahul	Delhi	India	Office Chair	Furniture
3	1004	2026-01-15	Sneha	Pune	India	Desk	Furniture
4	1005	2026-01-20	Amit	Bengaluru	India	Keyboard	Electronics

In [121…

```
## 16. CHECK EXCEL DATA
excel_df.shape

excel_df.info()

excel_df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24 entries, 0 to 23
Data columns (total 9 columns):
 #   Column      Non-Null Count  Dtype  
---  ------      --------------  -----  
 0   ORDER_ID    24 non-null     int64  
 1   ORDER_DATE  24 non-null     object 
 2   CUSTOMER    24 non-null     object 
 3   CITY        24 non-null     object 
 4   COUNTRY     24 non-null     object 
 5   PRODUCT     24 non-null     object 
 6   CATEGORY    24 non-null     object 
 7   QUANTITY    23 non-null     float64
 8   PRICE       23 non-null     float64
dtypes: float64(2), int64(1), object(6)
memory usage: 1.8+ KB
```

Out[121…

	0
ORDER_ID	0
ORDER_DATE	0
CUSTOMER	0
CITY	0
COUNTRY	0
PRODUCT	0
CATEGORY	0
QUANTITY	1
PRICE	1

dtype: int64

In [122…

```
## 17. LOAD JSON FILE
json_df = pd.read_json('sales.json')

json_df
```

Out[122…

	ORDER_ID	ORDER_DATE	CUSTOMER	CITY	COUNTRY	PRODUCT	CATEG
0	1001	2026-01-05	Amit	Pune	India	Laptop	Electro
1	1002	2026-01-08	Priya	Mumbai	India	Mouse	Electro
2	1003	2026-01-12	Rahul	Delhi	India	Office Chair	Furni
3	1004	2026-01-15	Sneha	Pune	India	Desk	Furni
4	1005	2026-01-20	Amit	Bengaluru	India	Keyboard	Electro
5	1006	2026-01-25	Priya	Mumbai	India	Monitor	Electro
6	1007	2026-02-02	Rahul	Delhi	India	Bookshelf	Furni
7	1008	2026-02-06	Neha	Pune	India	Tablet	Electro
8	1009	2026-02-10	Raj	Bengaluru	India	Sofa	Furni
9	1010	2026-02-14	Pooja	Mumbai	India	Headphones	Electro
10	1011	2026-02-18	Karan	Delhi	India	Printer	Electro
11	1012	2026-02-22	Riya	Pune	India	Cabinet	Furni
12	1013	2026-03-01	Arjun	Bengaluru	India	Laptop	Electro
13	1014	2026-03-05	Meera	Mumbai	India	Desk	Furni
14	1015	2026-03-10	Vikas	Delhi	India	Mouse	Electro
15	1016	2026-03-15	Anjali	Pune	India	Office Chair	Furni
16	1017	2026-03-20	Rohit	Bengaluru	India	Monitor	Electro
17	1018	2026-03-24	Kavya	Mumbai	India	Table	Furni
18	1019	2026-03-27	Sahil	Delhi	India	Keyboard	Electro
19	1020	2026-03-30	Nisha	Pune	India	Tablet	Electro
20	1021	2026-04-03	Varun	Mumbai	India	Printer	Electro
21	1022	2026-04-07	Isha	Delhi	India	Chair	Furni
22	1023	2026-04-10	Aarav	Pune	India	Laptop	Electro
23	1023	2026-04-10	Aarav	Pune	India	Laptop	Electro

In [123…

```
# 18. DISPLAY FIRST 5 JSON RECORDS
json_df.head()
```

Out[123…

	ORDER_ID	ORDER_DATE	CUSTOMER	CITY	COUNTRY	PRODUCT	CATEGORY
0	1001	2026-01-05	Amit	Pune	India	Laptop	Electronics
1	1002	2026-01-08	Priya	Mumbai	India	Mouse	Electronics
2	1003	2026-01-12	Rahul	Delhi	India	Office Chair	Furniture
3	1004	2026-01-15	Sneha	Pune	India	Desk	Furniture
4	1005	2026-01-20	Amit	Bengaluru	India	Keyboard	Electronics

In [124…

```
# 19. CHECK JSON DATA
json_df.shape

json_df.info()

json_df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24 entries, 0 to 23
Data columns (total 9 columns):
 #   Column      Non-Null Count  Dtype  
---  ------      --------------  -----  
 0   ORDER_ID    24 non-null     int64  
 1   ORDER_DATE  24 non-null     object 
 2   CUSTOMER    24 non-null     object 
 3   CITY        24 non-null     object 
 4   COUNTRY     24 non-null     object 
 5   PRODUCT     24 non-null     object 
 6   CATEGORY    24 non-null     object 
 7   QUANTITY    23 non-null     float64
 8   PRICE       23 non-null     float64
dtypes: float64(2), int64(1), object(6)
memory usage: 1.8+ KB
```

Out[124…

	0
ORDER_ID	0
ORDER_DATE	0
CUSTOMER	0
CITY	0
COUNTRY	0
PRODUCT	0
CATEGORY	0
QUANTITY	1
PRICE	1

dtype: int64

In [125…
```
# 20. COMPARE ALL THREE DATASETS
print("CSV Data Shape:", df.shape)

print("Excel Data Shape:", excel_df.shape)

print("JSON Data Shape:", json_df.shape)
```

```
CSV Data Shape: (24, 9)
Excel Data Shape: (24, 9)
JSON Data Shape: (24, 9)
```

In [126…
```
# 21. COMBINE DATASETS
all_data = pd.concat(
    [newdf, excel_df, json_df],
    ignore_index=True
)

all_data
```

Out[126…

	ORDER_ID	ORDER_DATE	CUSTOMER	CITY	COUNTRY	PRODUCT	CATEGOR
0	1001	2026-01-05	Amit	Pune	India	Laptop	Electronic
1	1002	2026-01-08	Priya	Mumbai	India	Mouse	Electronic
2	1003	2026-01-12	Rahul	Delhi	India	Office Chair	Furnitur
3	1004	2026-01-15	Sneha	Pune	India	Desk	Furnitur
4	1005	2026-01-20	Amit	Bengaluru	India	Keyboard	Electronic
…	…	…	…	…	…	…	
66	1020	2026-03-30	Nisha	Pune	India	Tablet	Electronic
67	1021	2026-04-03	Varun	Mumbai	India	Printer	Electronic
68	1022	2026-04-07	Isha	Delhi	India	Chair	Furnitur
69	1023	2026-04-10	Aarav	Pune	India	Laptop	Electronic
70	1023	2026-04-10	Aarav	Pune	India	Laptop	Electronic

71 rows × 10 columns

In [127…
```
# 22. CHECK COMBINED DATA
all_data.shape

all_data.head()

all_data.tail()
```

Out[127…

	ORDER_ID	ORDER_DATE	CUSTOMER	CITY	COUNTRY	PRODUCT	CATEGORY
66	1020	2026-03-30	Nisha	Pune	India	Tablet	Electronics
67	1021	2026-04-03	Varun	Mumbai	India	Printer	Electronics
68	1022	2026-04-07	Isha	Delhi	India	Chair	Furniture
69	1023	2026-04-10	Aarav	Pune	India	Laptop	Electronics
70	1023	2026-04-10	Aarav	Pune	India	Laptop	Electronics

In [128…

```
# 23. CHECK MISSING VALUES
all_data.isnull().sum()
```

Out[128…

	0
ORDER_ID	0
ORDER_DATE	0
CUSTOMER	0
CITY	0
COUNTRY	0
PRODUCT	0
CATEGORY	0
QUANTITY	2
PRICE	2
LOCATION	48

dtype: int64

In [129…

```
# 24. FILL MISSING VALUES
all_data = all_data.fillna(0)

all_data.isnull().sum()
```

Out[129…

	0
ORDER_ID	0
ORDER_DATE	0
CUSTOMER	0
CITY	0
COUNTRY	0
PRODUCT	0
CATEGORY	0
QUANTITY	0
PRICE	0
LOCATION	0

dtype: int64

In [130…
```
# 25. REMOVE DUPLICATES
all_data = all_data.drop_duplicates()

all_data.shape
```

Out[130… (46, 10)

In [131…
```
# 26. DATA TRANSFORMATION
# Create Total Sales column
all_data['TOTAL_SALES'] = (
    all_data['QUANTITY'] * all_data['PRICE']
)

all_data.head()
```

Out[131…

	ORDER_ID	ORDER_DATE	CUSTOMER	CITY	COUNTRY	PRODUCT	CATEGORY
0	1001	2026-01-05	Amit	Pune	India	Laptop	Electronics
1	1002	2026-01-08	Priya	Mumbai	India	Mouse	Electronics
2	1003	2026-01-12	Rahul	Delhi	India	Office Chair	Furniture
3	1004	2026-01-15	Sneha	Pune	India	Desk	Furniture
4	1005	2026-01-20	Amit	Bengaluru	India	Keyboard	Electronics

In [132…
```
# 27. CREATE LOCATION COLUMN
all_data['LOCATION'] = (
    all_data['CITY'] + ',' + all_data['COUNTRY']
)

all_data.head()
```

Out[132…

	ORDER_ID	ORDER_DATE	CUSTOMER	CITY	COUNTRY	PRODUCT	CATEGORY
0	1001	2026-01-05	Amit	Pune	India	Laptop	Electronics
1	1002	2026-01-08	Priya	Mumbai	India	Mouse	Electronics
2	1003	2026-01-12	Rahul	Delhi	India	Office Chair	Furniture
3	1004	2026-01-15	Sneha	Pune	India	Desk	Furniture
4	1005	2026-01-20	Amit	Bengaluru	India	Keyboard	Electronics

In [133…
```
# 28. TOTAL SALES
total_sales = all_data['TOTAL_SALES'].sum()

print("Total Sales =", total_sales)
```

```
Total Sales = 952400.0
```

In [134…
```
# 29. AVERAGE SALES
average_sales = all_data['TOTAL_SALES'].mean()

print("Average Sales =", average_sales)
```

```
Average Sales = 20704.347826086956
```

In [135…
```
# 30. MAXIMUM SALES
maximum_sales = all_data['TOTAL_SALES'].max()

print("Maximum Sales =", maximum_sales)
```

```
Maximum Sales = 110000.0
```

In [136…
```
# 31. MINIMUM SALES
minimum_sales = all_data['TOTAL_SALES'].min()

print("Minimum Sales =", minimum_sales)
```

```
Minimum Sales = 0.0
```

In [137…
```
# 32. TOTAL QUANTITY SOLD
total_quantity = all_data['QUANTITY'].sum()

print("Total Quantity Sold =", total_quantity)
```

```
Total Quantity Sold = 92.0
```

In [138…
```
# 33. CATEGORY-WISE SALES
category_sales = all_data.groupby(
    'CATEGORY'
)['TOTAL_SALES'].sum()

category_sales
```

Out[138…

CATEGORY	TOTAL_SALES
Electronics	747200.0
Furniture	205200.0

dtype: float64

In [139…

```
# 34. PRODUCT-WISE SALES
product_sales = all_data.groupby(
    'PRODUCT'
)['TOTAL_SALES'].sum()

product_sales
```

Out[139…

PRODUCT	TOTAL_SALES
Bookshelf	13000.0
Cabinet	28000.0
Chair	0.0
Desk	46000.0
Headphones	20000.0
Keyboard	15400.0
Laptop	456000.0
Monitor	75000.0
Mouse	14800.0
Office Chair	43200.0
Printer	30000.0
Sofa	56000.0
Table	19000.0
Tablet	136000.0

dtype: float64

In [140…

```
# 35. CITY-WISE SALES
city_sales = all_data.groupby(
    'CITY'
)['TOTAL_SALES'].sum()

city_sales
```

Out[140…

CITY	TOTAL_SALES
Bengaluru	212000.0
Delhi	75200.0
Mumbai	124000.0
Pune	541200.0

dtype: float64

In [141…

```
# 36. COUNTRY-WISE SALES
country_sales = all_data.groupby(
    'COUNTRY'
)['TOTAL_SALES'].sum()

country_sales
```

Out[141…

COUNTRY	TOTAL_SALES
India	952400.0

dtype: float64

In [142…

```
# 37. SORT CATEGORY-WISE SALES
category_sales.sort_values(
    ascending=False
)
```

Out[142…

CATEGORY	TOTAL_SALES
Electronics	747200.0
Furniture	205200.0

dtype: float64

In [143…

```
# 38. SORT PRODUCT-WISE SALES
product_sales.sort_values(
    ascending=False
)
```

Out[143…

PRODUCT	TOTAL_SALES
Laptop	456000.0
Tablet	136000.0
Monitor	75000.0
Sofa	56000.0
Desk	46000.0
Office Chair	43200.0
Printer	30000.0
Cabinet	28000.0
Headphones	20000.0
Table	19000.0
Keyboard	15400.0
Mouse	14800.0
Bookshelf	13000.0
Chair	0.0

dtype: float64

In [144…

```
# 39. HIGHEST SELLING PRODUCT
product_sales.sort_values(
    ascending=False
).head(1)
```

Out[144…

PRODUCT	TOTAL_SALES
Laptop	456000.0

dtype: float64

In [145…

```
# 40. HIGHEST SELLING CATEGORY
category_sales.sort_values(
    ascending=False
).head(1)
```

Out[145…

CATEGORY	TOTAL_SALES
Electronics	747200.0

dtype: float64

In [146…

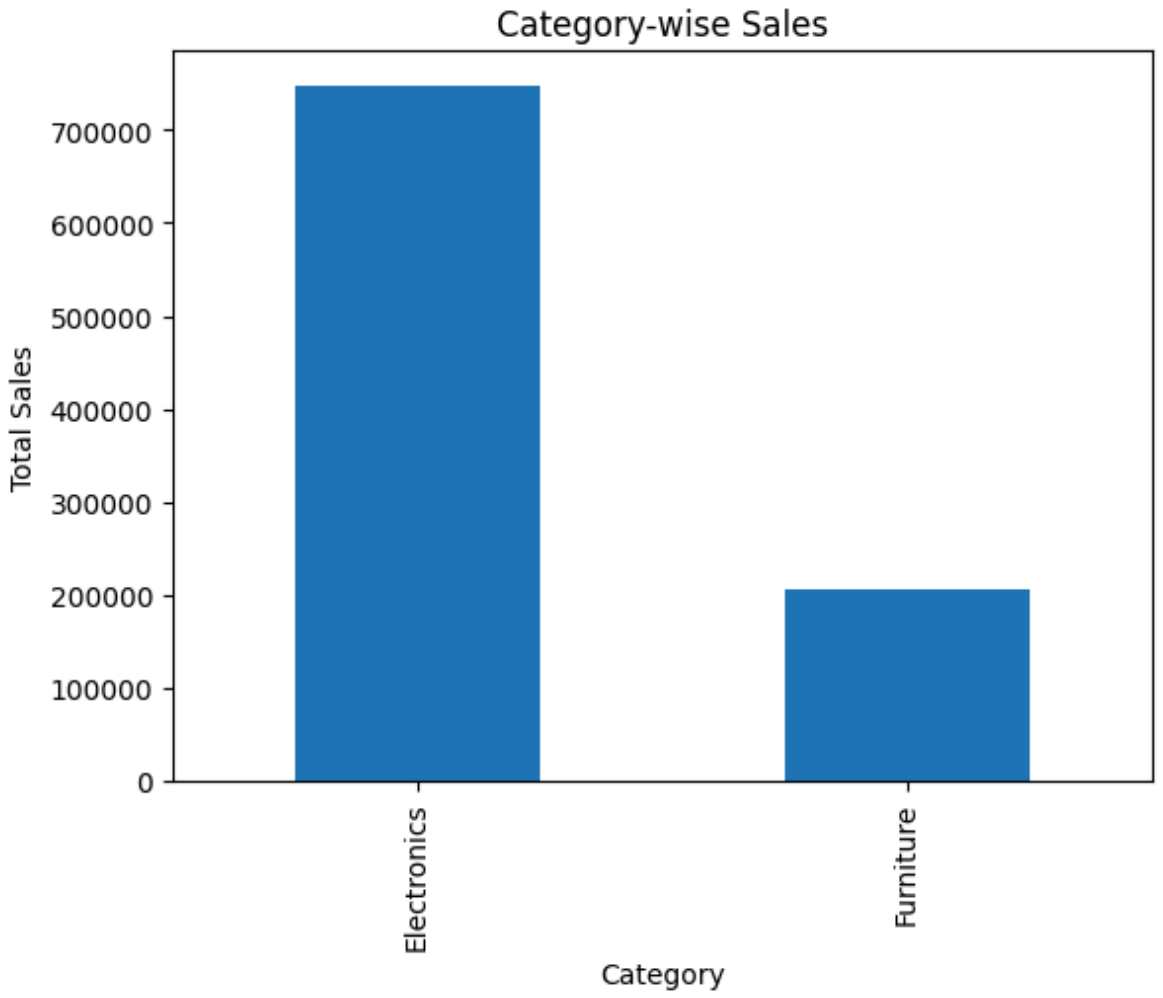
```
# 41. BAR PLOT - CATEGORY-WISE SALES
category_sales.plot(
    kind='bar'
)

plt.title('Category-wise Sales')

plt.xlabel('Category')

plt.ylabel('Total Sales')

plt.show()
```



In [147…

```
# 42. BAR PLOT - PRODUCT-WISE SALES
product_sales.plot(
    kind='bar'
)

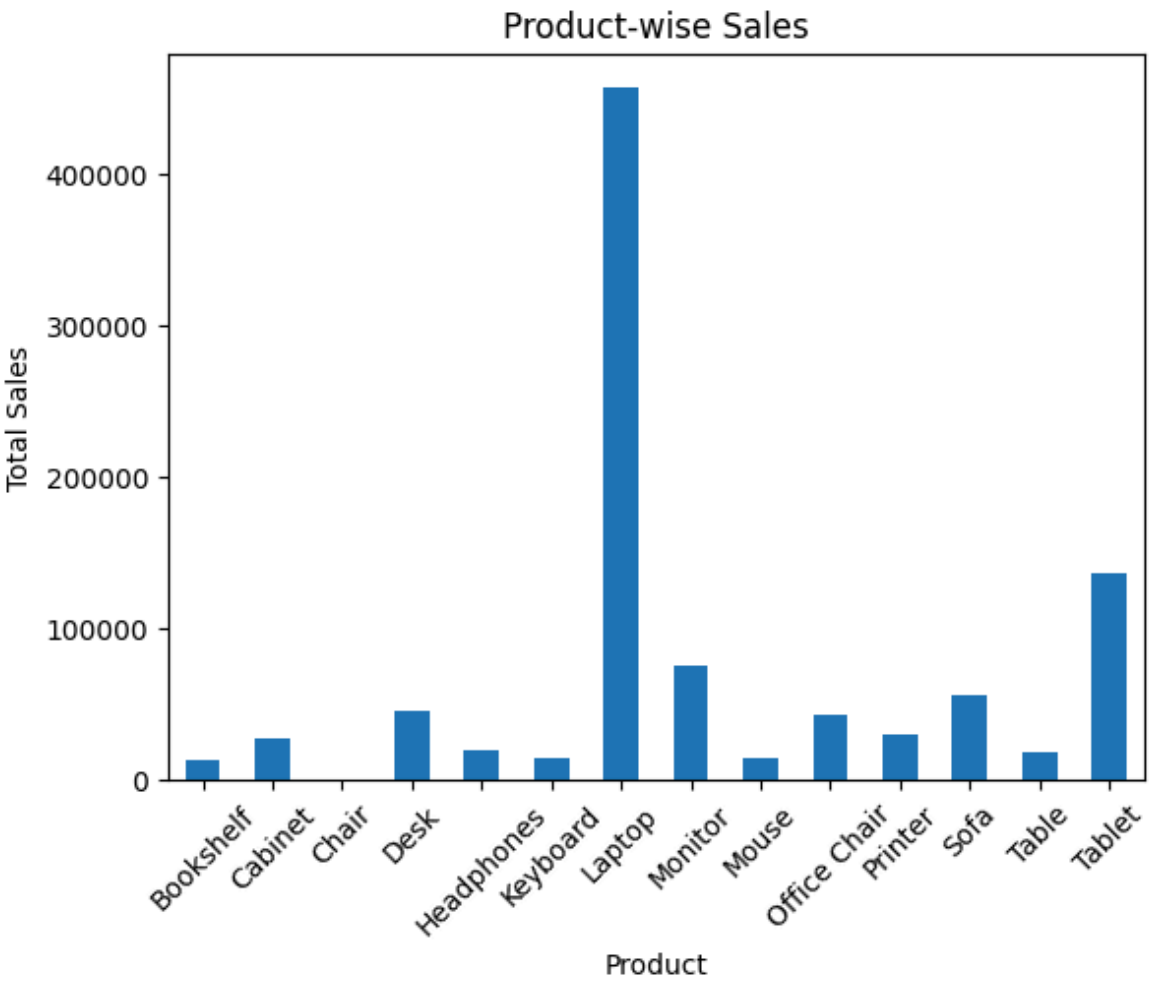
plt.title('Product-wise Sales')
```

```
plt.xlabel('Product')

plt.ylabel('Total Sales')

plt.xticks(rotation=45)

plt.show()
```



In [148…

```
# 43. BAR PLOT - CITY-WISE SALES
city_sales.plot(
    kind='bar'
)

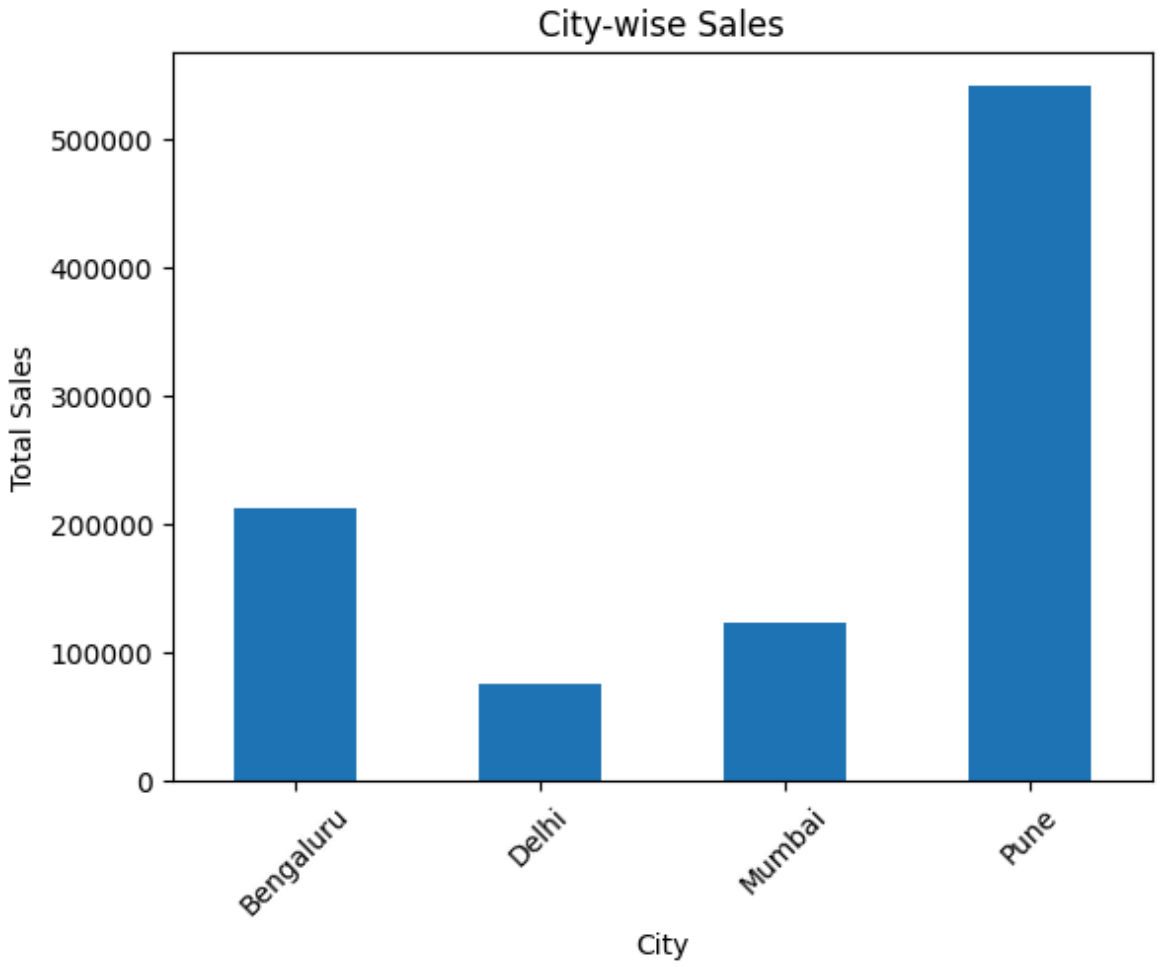
plt.title('City-wise Sales')

plt.xlabel('City')

plt.ylabel('Total Sales')

plt.xticks(rotation=45)

plt.show()
```



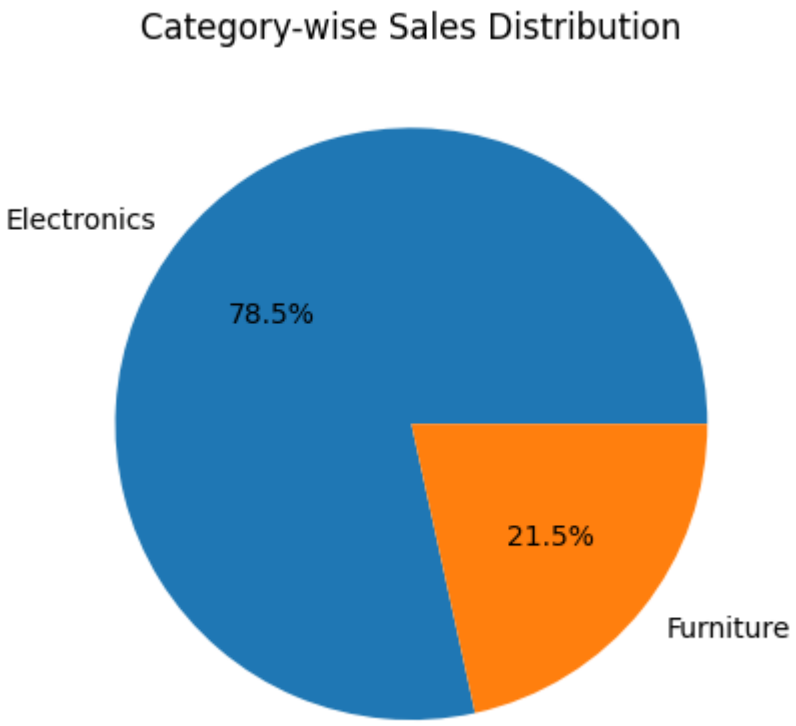
In [149…

```
# 44. PIE CHART - CATEGORY-WISE SALES
category_sales.plot(
    kind='pie',
    autopct='%1.1f%%'
)

plt.title('Category-wise Sales Distribution')

plt.ylabel('')

plt.show()
```



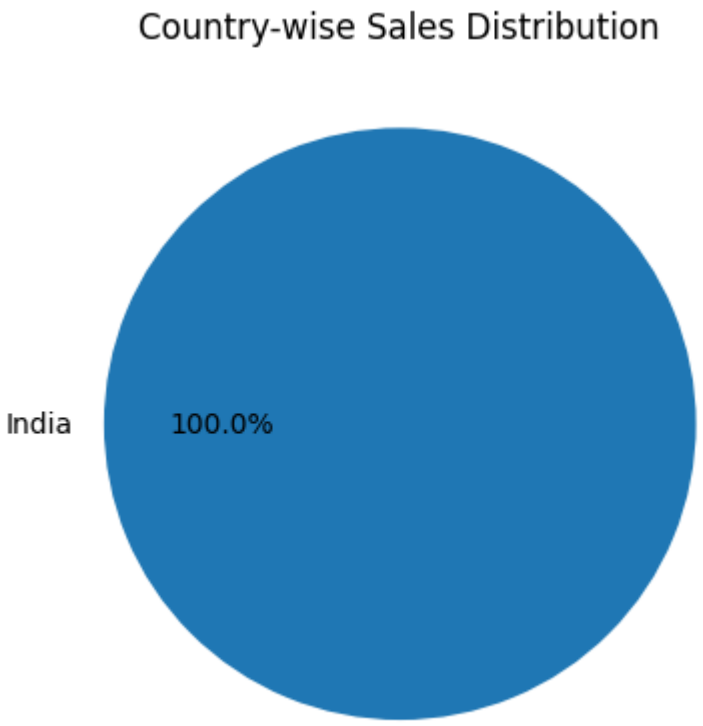
In [150…

```
# 45. PIE CHART - COUNTRY-WISE SALES
country_sales.plot(
    kind='pie',
    autopct='%1.1f%%'
)

plt.title('Country-wise Sales Distribution')

plt.ylabel('')

plt.show()
```



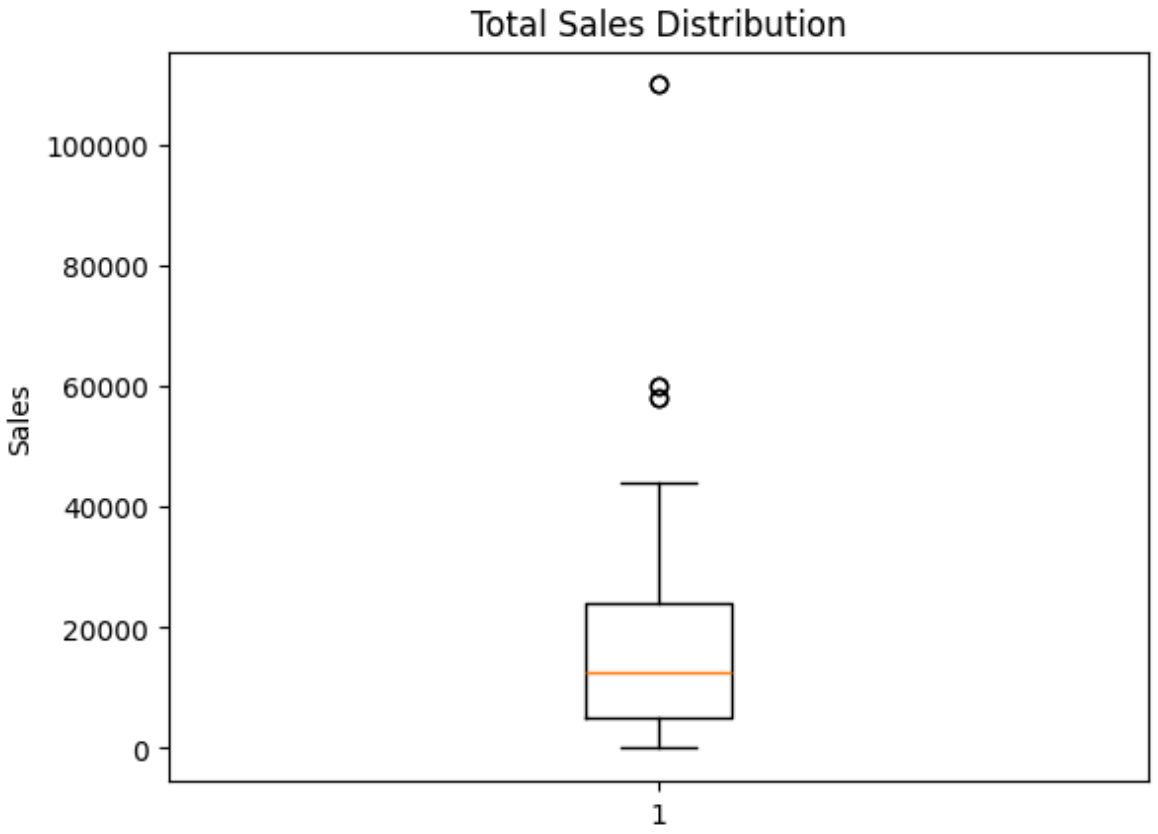
In [151…

```
# 46. BOX PLOT - TOTAL SALES
plt.boxplot(
    all_data['TOTAL_SALES']
)

plt.title('Total Sales Distribution')

plt.ylabel('Sales')

plt.show()
```



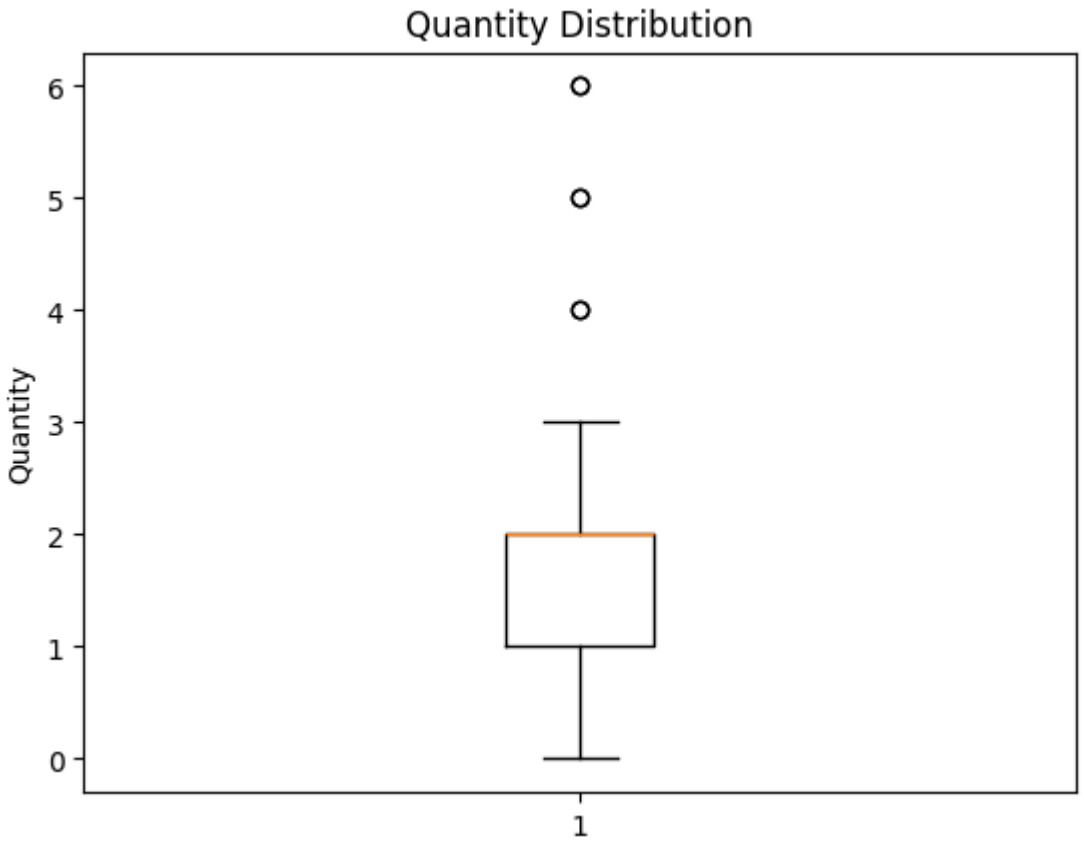
In [152…

```
# 47. BOX PLOT - QUANTITY
plt.boxplot(
    all_data['QUANTITY']
)

plt.title('Quantity Distribution')

plt.ylabel('Quantity')

plt.show()
```



In [153…
```python
# 48. SAVE CLEANED DATA AS CSV
all_data.to_csv(
    'cleaned_sales.csv',
    index=False
)

print("cleaned_sales.csv created successfully")
```

```
cleaned_sales.csv created successfully
```

In [154…
```python
# 49. SAVE CLEANED DATA AS EXCEL
all_data.to_excel(
    'cleaned_sales.xlsx',
    index=False
)

print("cleaned_sales.xlsx created successfully")
```

```
cleaned_sales.xlsx created successfully
```

In [155…
```python
# 50. SAVE CLEANED DATA AS JSON
all_data.to_json(
    'cleaned_sales.json',
    orient='records',
    indent=4
)

print("cleaned_sales.json created successfully")
```

```
cleaned_sales.json created successfully
```

In [155…